Case Study: Noida Library System

Noida Public Library is the biggest library in Noida. Currently it has about 300 members. A person who is 18 or above can become a member. There is a membership fee of Rs 400 for a year. There is a form to be filled in which person fills personal details. These forms are kept in store for maintaining members' records and knowing the membership period.

A member can issue a maximum of three books. He/she has three cards to issue books. Against each card a member can issue one book from library. Whenever a member wishes to issue a book and there are spare cards, then the book is issued. Otherwise that request is not entertained. Each book is to be returned on the specified due date. If a member fails to return a book on the specified date, a fine of Rs 2 per day after the due return date is charged. If in case a card gets lost then a duplicate card is issued. Accounts are maintained for the membership fees and money collected from the fines. There are two librarians for books return and issue transaction. Approximately 100 members come to library daily to issue and return books.

There are 5000 books available out of which 1000 books are for reference and cannot be issued.

Records for the books in the library are maintained. These records contain details about the publisher, author, subject, language, etc. There are suppliers that supply books to the library. Library maintains records of these suppliers.

Many reports are also produced. These reports are for details of the books available in the library, financial details, members' details, and supplier's details.

Currently all functions of the library are done manually. Even the records are maintained on papers.

Now day by day members are increasing. Maintaining manual records is becoming difficult task.

There are other problems also that the library staff is facing. Like in case of issue of duplicate cards to a member when member or library staff loses the card. It is very difficult to check the genuinity of the problem.

Sometimes the library staff needs to know about the status of a book as to whether it is issued or not.

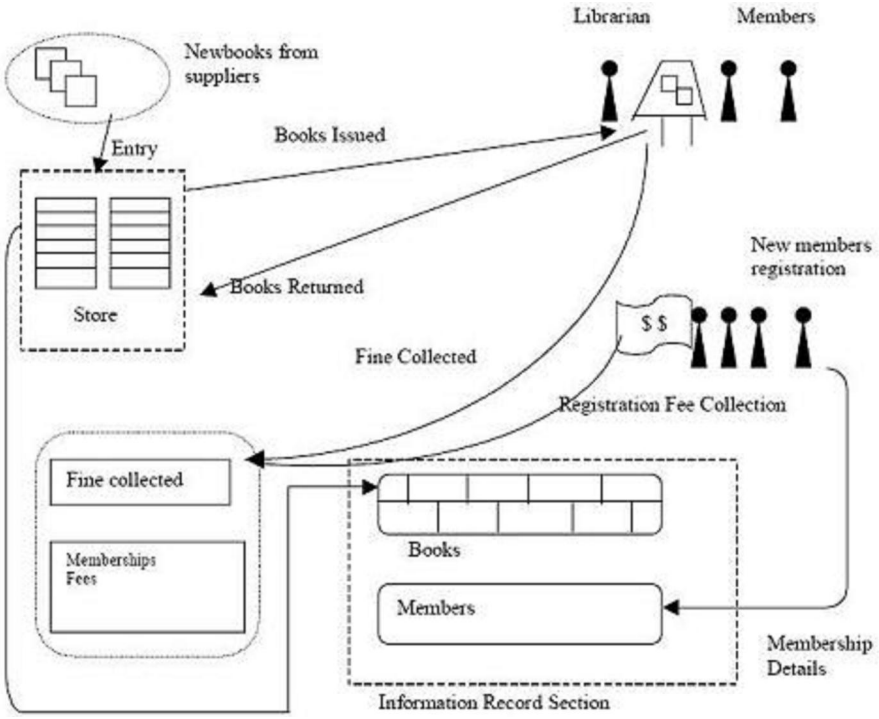
So to perform this kind of search is very difficult in a manual system.

Also management requires reports for books issued, books in the library, members, and accounts.

Manually producing the reports is a cumbersome job when there are hundreds and thousands of records.

Management plans to expand the library, in terms of books, number of members and finally the revenue generated. It is observed that every month there are at least 50-100 requests for membership. For the last two months the library has not entertained requests for the new membership as it was difficult to manage the existing 250 members manually. With the expansion plans, the management of the library aims to increase its members at the rate of 75 per month. It also plans to increase the membership fees from 400 to 1000 for yearly and 500 for half year, in order to provide its members better services, which includes increase in number of books from 3 to 4.

Due to the problems faced by the library staff and its expansion plans, the management is planning to have a system that would first eradicate the needs of cards. A system to automate the functions of record keeping and report generation. And which could help in executing the different searches in a faster manner. The system to handle the financial details.



Applying the concepts studied to the case study:

The first thing we studied is systems. In our case study Noida Public Library is our system. Every system is a set of some functional units that work together to achieve some objective. The main objective of library system is to provide books to its members without difficulty. The figure above depicts our library system pictorially.

Our system has many functional units. Books issue and return section, books record unit, members record unit, accounts, and report generation units are the different functional units of the library. Each functional unit has its own task. However, each of these work independently to achieve the overall objective of the library.

Later in the session, we talked about different components and characteristics of the systems. Data is an important component of any system. Here, data is pertaining to the details of members, books, accounts, and suppliers. Since people can interact with the system this system is an open system.

The system is mainly concerned with the management of data it is an information system.

If this system were to be automated as conceived by the management, then role of the system analyst would be to study the system, its workings, and its existing problems. Also the analyst needs to provide a solution to the existing problem.

Now that the management has decided for an automated system the analyst would perform the above tasks. As the analyst did the study of the system, the following problems were identified

- Maintaining membership cards
- Producing reports due to large amount of data
- Maintaining accounts
- Keeping records for books in library and its members
- Performing searches

Now that the analyst has studied the system and identified the problems, it is the responsibility of the analyst to provide a solution system to the management of the library.